

基於生成式 AI 的瑕疵資料生成 與錒錫瑕疵檢測方法之研究

Leveraging Generative AI for Defect Data Generation and
Solder Defect Detection in PCB Manufacturing

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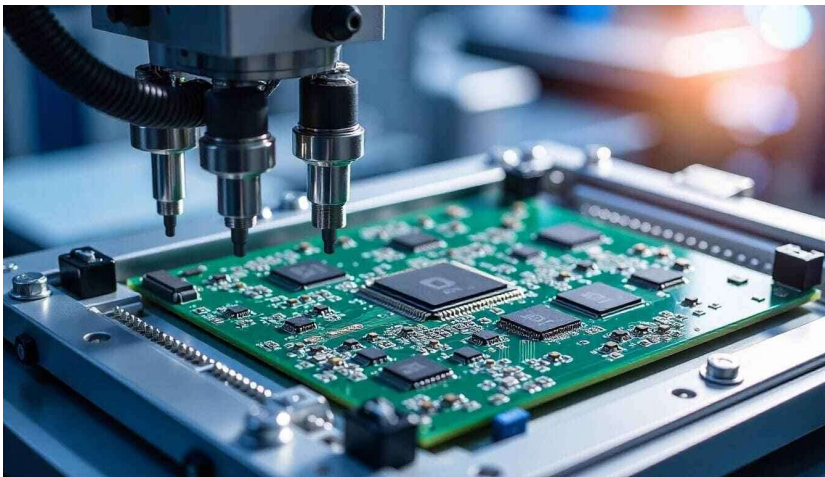
王尉任 博士

林家瑜 博士

Introduction

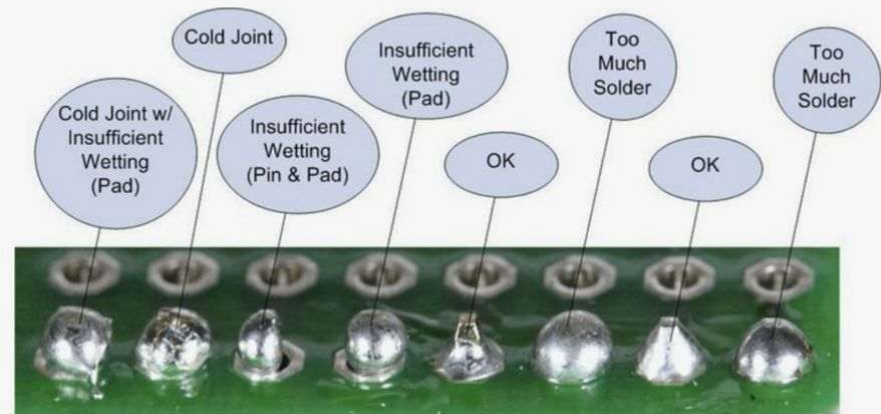
Background

- In PCB manufacturing, a large number of tiny components and solder joints are densely distributed across the board. After the soldering process, various types of soldering defects can easily occur, such as short circuits, missing solder, excessive solder, and insufficient solder.
- These defects may only affect the product appearance in mild cases, but in severe cases, they can **lead to circuit malfunction or even complete failure**.



<https://www.lorric.com/tw/Industries/Printed-Circuit-Board-PCB>

Common Types of PTH Assembly Defects



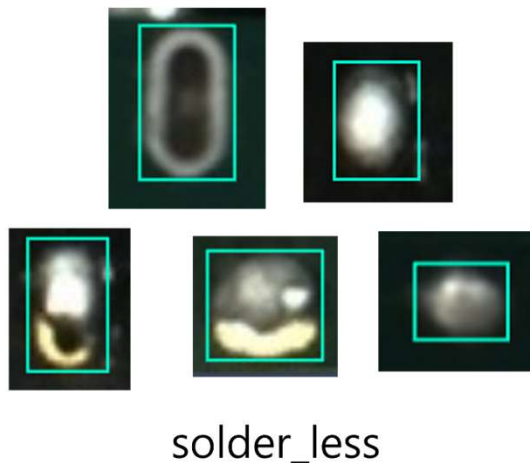
<https://www.viasion.com/blog/how-to-check-defects-in-pcb-soldering>

Background

- Traditionally, defect inspection relies on manual visual checks of each board. While this approach can be accurate to some extent, it is time-consuming and labor-intensive. When inspecting densely packed and tiny solder joints over long periods, human operators are prone to fatigue, which increases the risk of missed detections.
- With the rapid advancement of technology, various deep learning models, such as ResNet, Faster R-CNN, and YOLO, are now available for automated defect detection.
- However, the effectiveness of these models highly depends on the quality and diversity of the training data. **Sufficient and varied defect samples** are essential to ensure both high accuracy and good generalization in real-world applications.

Background

- In practice, collecting high-quality defect data is challenging. PCB production lines usually maintain a high yield rate, which means that **actual defect samples are rare**.
- Moreover, each defect type can exhibit a **wide range of visual patterns**, and the annotation process is both time-consuming and labor-intensive.
- As a result, image datasets in industrial environments are often **limited in size and imbalanced across classes**, negatively impacting the training and performance of deep learning models.



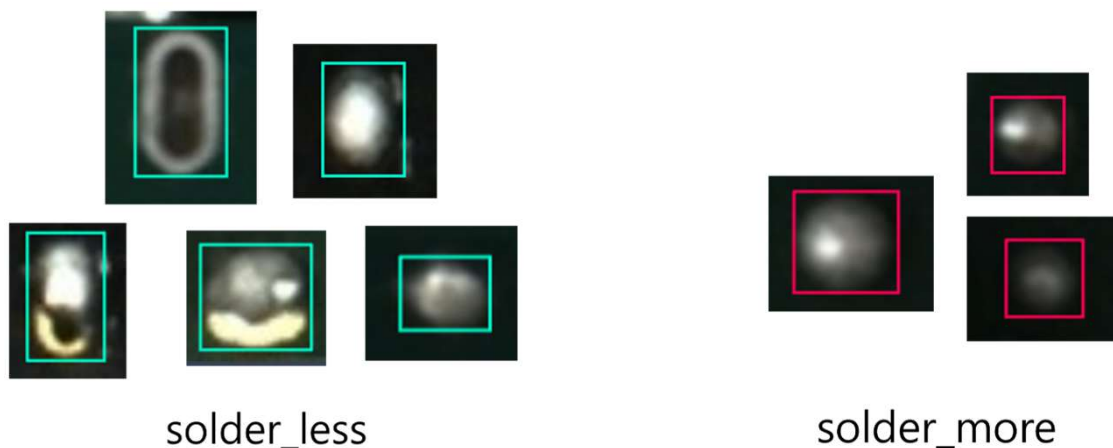
Motivation

- The quality and quantity of training data play a critical role in the performance of deep learning models. However, data scarcity remains a widespread issue across many domains, especially in industrial and medical applications, where data is often sensitive and difficult to obtain.

	traditional	generative model-based	diffusion model-based
advantages	rapidly increase data diversity, highly flexible in use	high variability	high-resolution, realistic images
disadvantages	limited variability	requires a large and diverse dataset for training	generated data cannot match the desired characteristics

Motivation

- In addition, the dataset used in our study contains two types of visually similar defects: solder_less and solder_more. In practice, these defects are typically distinguished using 3D imaging and modeling techniques, which allow for accurate measurement of solder volume. However, **3D equipment is costly**, and most manufacturing facilities still **rely primarily on 2D inspection systems**.
- Therefore, this study aims to leverage the generative capabilities of diffusion models to **address the problem of insufficient PCB defect data** by generating high-fidelity synthetic images that capture subtle differences between visually similar defect types, thereby enhancing the accuracy of defect detection models in distinguishing such categories.



Problem Definition

- How existing diffusion models can be applied in the context of PCB defect detection to **generate diverse and realistic defect images** under the constraints of **using only 2D visual data and limited training samples**, in order to **improve the accuracy of defect detection models**.
- The quality of the generated images is evaluated using **Fréchet Inception Distance (FID)**, while the overall performance of defect detection is assessed using standard metrics including **Precision, Recall, their class-wise and average values**, as well as **mean Average Precision at IoU threshold 0.5 (mAP@50)**.

Contribution

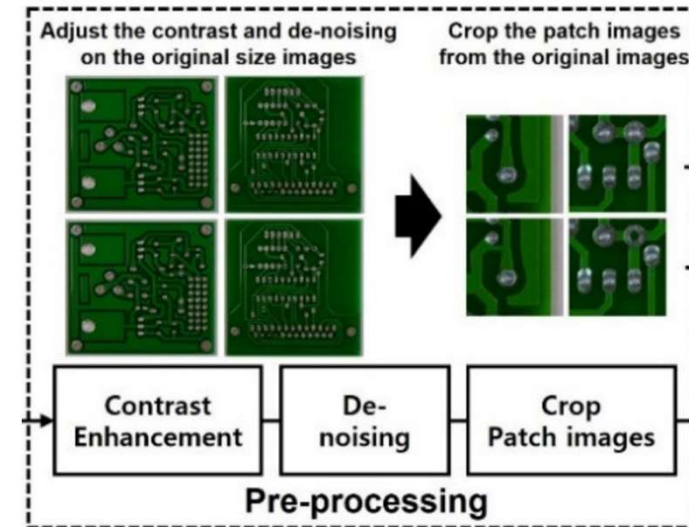
- The proposed method in this study is capable of generating defect PCB images that **closely resemble real samples** while introducing **meaningful variations in solder joint details**. This **address the common issue of insufficient training data in deep learning tasks**.
- By combining the generated data with original samples for fine-tuning the YOLOv11s model, we observed a notable improvement in overall detection performance.

Metric	Improve
mean precision	0.036(3.9%)
mean recall	0.045(5.2%)
solder_more precision/recall	0.012(1.2%)/0.183(22.5%)
solder_less precision/recall	-0.012(-1.5%)/0.03(3.41%)

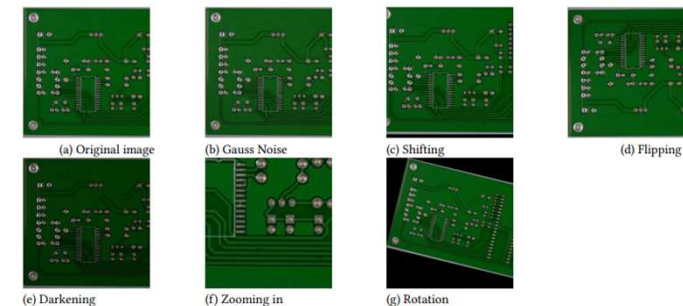
Related works

Traditional Data Augmentation

- In 2021, Jungsuk Kim et al. proposed a PCB defect detection method based on deep learning[1].
- In 2023, Xiangyuan Zhu et al. conducted a study on automated PCB defect detection[2].
- In 2022, Abhiroop Bhattacharya et al. proposed a deep learning framework for PCB defect classification[3]. When they removed the default data augmentation module from YOLOv5s, the mAP@50 dropped significantly from 94.5 to 84.7.



data preprocessing in the method proposed by Jungsuk Kim et al.



data augmentation in the method proposed by Xiangyuan Zhu et al.

- [1] J. Kim, J. Ko, H. Choi, and H. Kim, "Printed circuit board defect detection using deep learning via a skip-connected convolutional autoencoder," *Sensors*, vol. 21, no. 15, p. 4968, 2021.
- [2] X. Zhu, X. Xiao, Z. Lan, Q. Hong, and M. Hou, "An intelligent defect detection algorithm for pcb based on deep learning," in *Proceedings of the 6th International Conference on Advances in Image Processing*, pp. 20–25, 2022.
- [3] A. Bhattacharya and S. G. Cloutier, "End-to-end deep learning framework for printed circuit board manufacturing defect classification," *Scientific reports*, vol. 12, no. 1, p. 12559, 2022.

Generative Model-Based Data Augmentation

- In 2023, Chenglong Wang et al. proposed a data augmentation approach that **combines Conditional GAN and TransGAN**[1]. Their experiments showed that incorporating generated data **improved mAP by 1.1% to 2.7%** across various object detection models compared to using only original data.
- In 2023, Chang-Yeh Hsieh et al. introduced a data augmentation method[2] based on Wasserstein GAN (WGAN). Their results demonstrated that applying WGAN-based augmentation increased the test set **accuracy from 98.78% to 99.12%**.

[1] C. Wang, G. Huang, Z. Huang, and W. He, "Conditional transgan-based data augmentation for pcb electronic component inspection," Computational Intelligence and Neuroscience, vol. 2023, no. 1, p. 2024237, 2023.

[2] C.-Y. Hsieh, L.-S. Tseng, Y.-H. Chen, C.-H. Tsai, and C.-H. Wu, "Deep-learning methods for defect inspection of plated through holes with clustering-based auto-labeling and gan-based model training," IEEE Access, 2024.

Diffusion Model-Based Data Augmentation

- In 2024, Federico Girella et al. proposed a training-free data augmentation approach[1] for industrial surface defect detection using Latent Diffusion Models, known as DIAG.
- In 2025, Youzhi Xu et al. introduced a method[2] that **combines ControlNet with Latent Diffusion Models** for augmenting PCB defect data. This approach was specifically designed to address the issue of data scarcity in PCB applications.

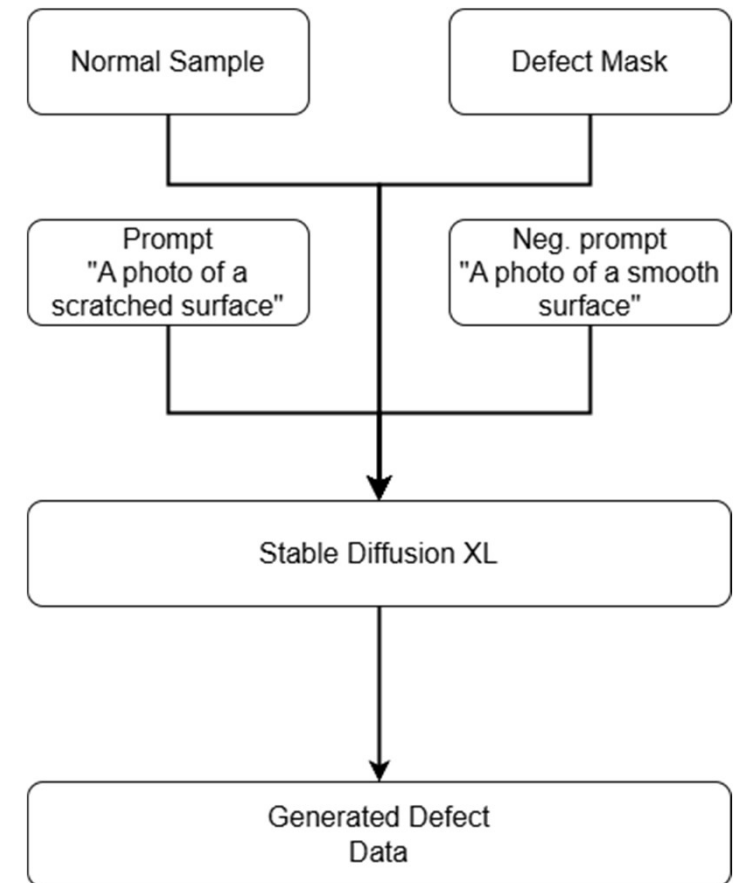
[1] F. Girella, Z. Liu, F. Fummi, F. Setti, M. Cristani, and L. Capogrosso, "Leveraging latent diffusion models for training-free in-distribution data augmentation for surface defect detection," in 2024 International Conference on Content-Based Multimedia Indexing (CBMI), pp. 1–7, IEEE, 2024.

[2] Y. Xu, H. Wu, Y. Liu, and X. Liu, "Printed circuit board sample expansion and automatic defect detection based on diffusion models and convnext," Micromachines, vol. 16, no. 3, p. 261, 2025.

DIAG

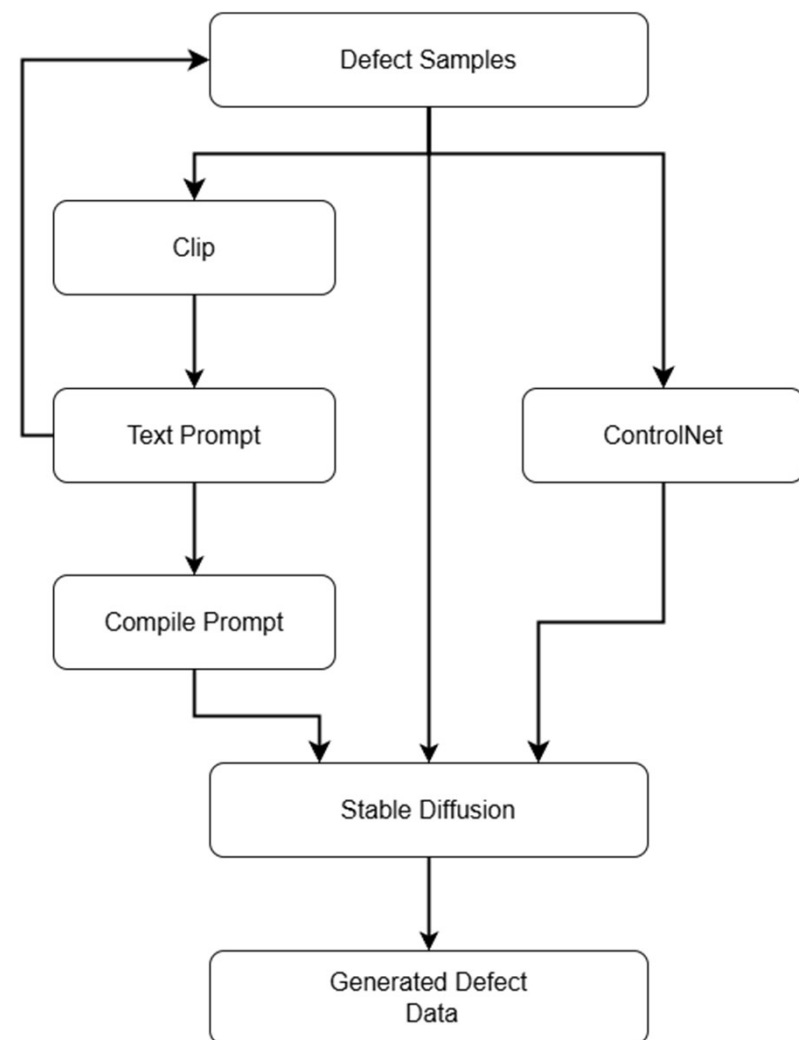
1. Select a defect-free surface image as the base image.
2. Randomly select a defect mask from the defect dataset to **simulate the location and shape** of a defect in a real sample.
3. Input the base image, defect mask, along with the corresponding positive and negative prompts into a Latent Diffusion Model (Stable Diffusion XL) to synthesize defect images.

positive prompt : “white marks on the wall” and “copper metal scratches”
 negative prompt: “smooth, plain, black, dark, shadow”



CSDG

1. Use CLIP to generate a caption (text description) for each image.
2. Aggregate all captions and extract the most frequently occurring words to form the prompt.
3. Use ControlNet to extract image features such as Canny edges and depth maps.
4. Input the prompt, defect data, and extracted features into Stable Diffusion V1.5 to generate synthetic defect images.

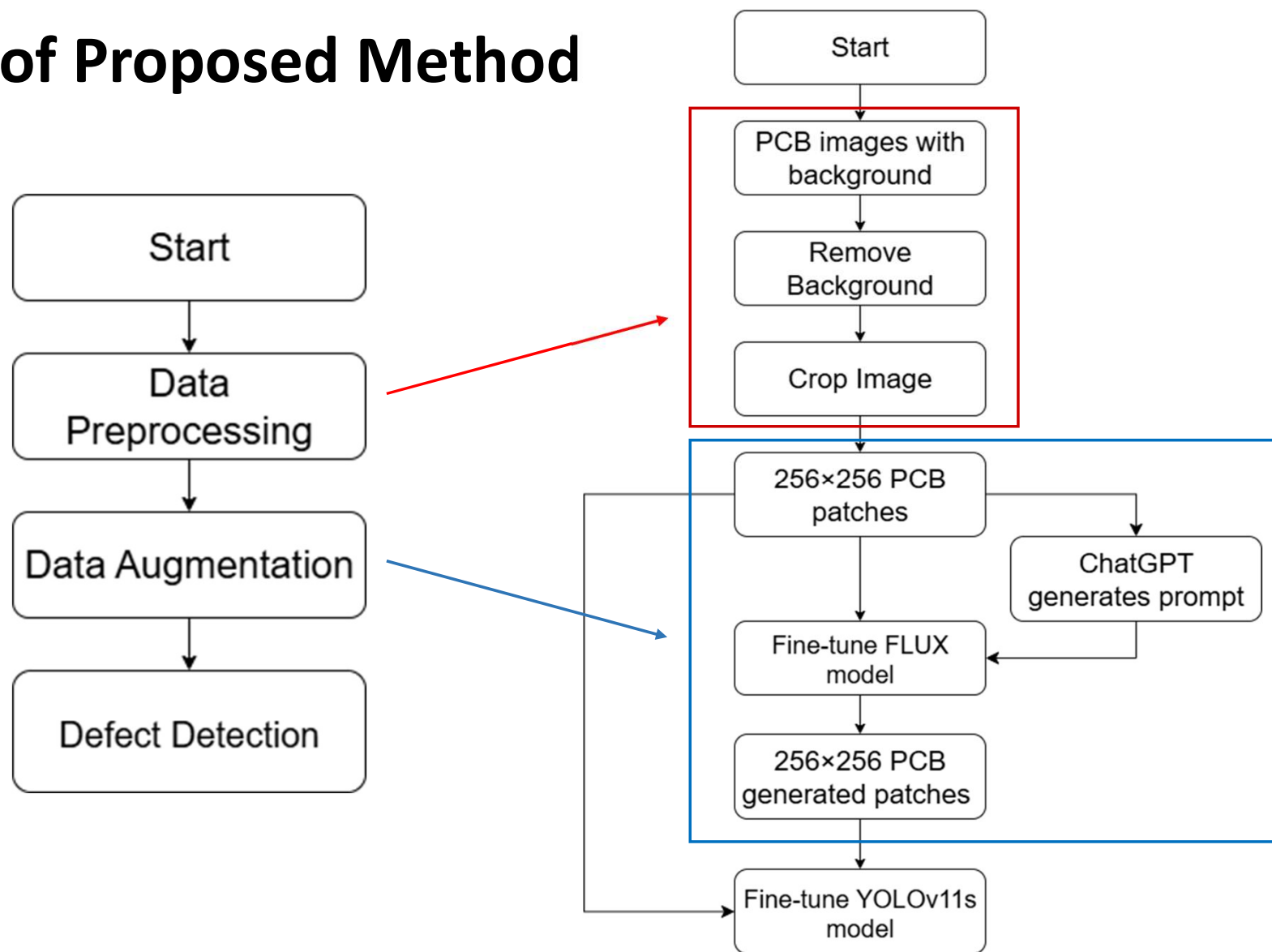


Method Comparison

	prompt tailored to the dataset	generate the entire image	image details are distorted
DIAG	×	×	×
CSDG	✓	✓	×
Proposed	✓	✓	✓

Proposed Method

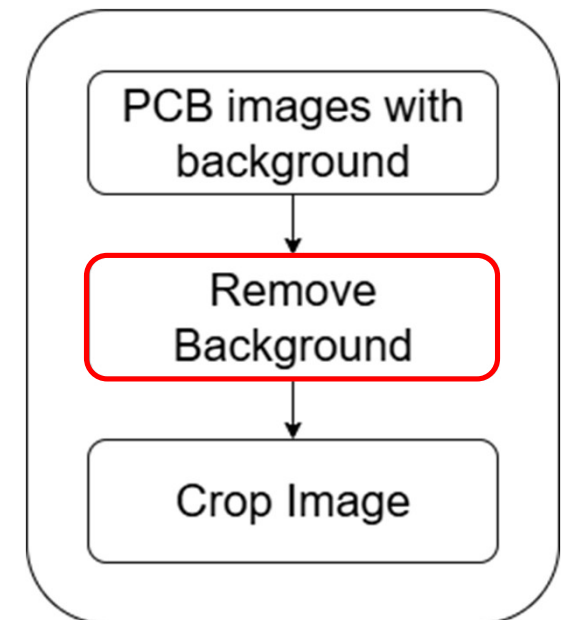
Flowchart of Proposed Method



Data Preprocessing - Remove Background

- Original images contain various types of PCB boards with inconsistent sizes.
- Adopt the Language Segment Anything[1] tool for target region segmentation. This tool is built upon the Segment Anything[2] framework and enables semantic-guided segmentation using text prompts.
- In our dataset, using the prompt "PCB" allows accurate segmentation of the PCB regions. The bounding box of the segmented result is then extracted and used as a cropping reference to isolate the main PCB area. This ensures consistency and facilitates subsequent processing and analysis.

Data Preprocessing



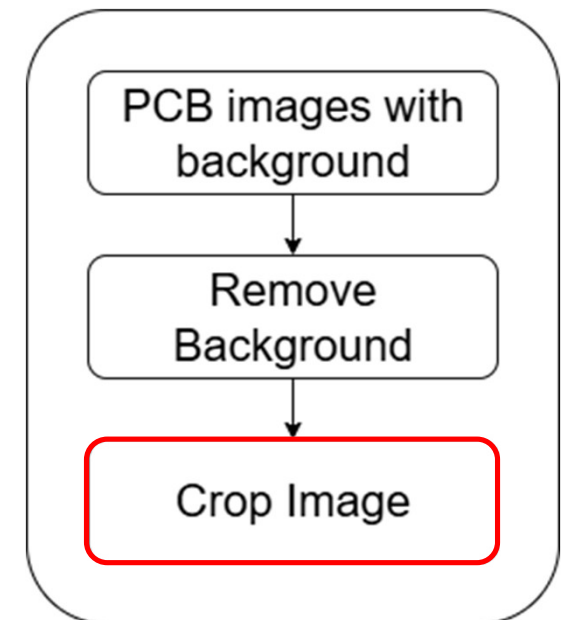
[1] luca medeiros, "Language Segment-Anything: Sam with text prompt." <https://github.com/lucamedeiros/lang-segment-anything>, 2024. GitHub repository, accessed: 2025-06-10.

[2] X. Xu, Z. Wang, G. Zhang, K. Wang, and H. Shi, "Versatile diffusion: Text, images and variations all in one diffusion model," in Proceedings of the IEEE/CVF International Conference on ComputerVision, pp. 7754–7765, 2023.

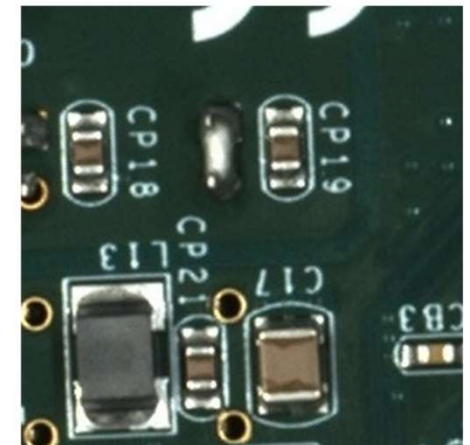
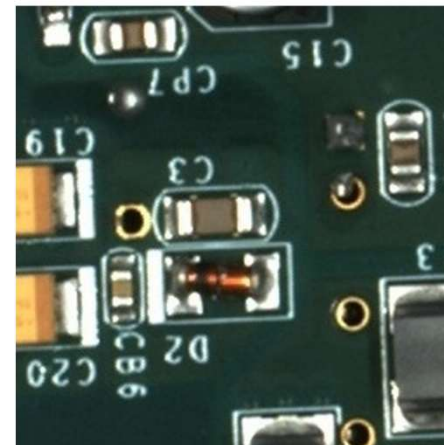
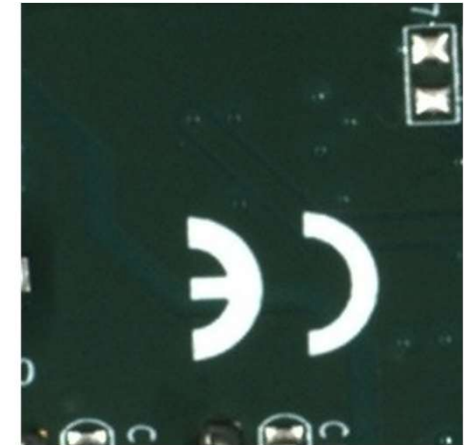
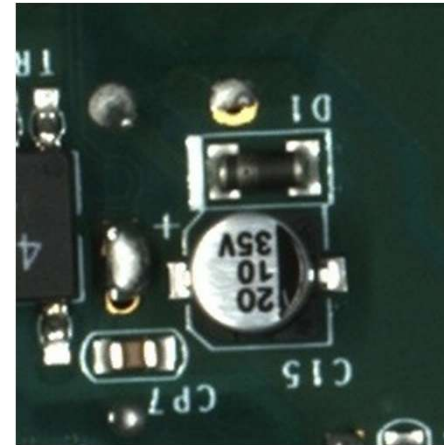
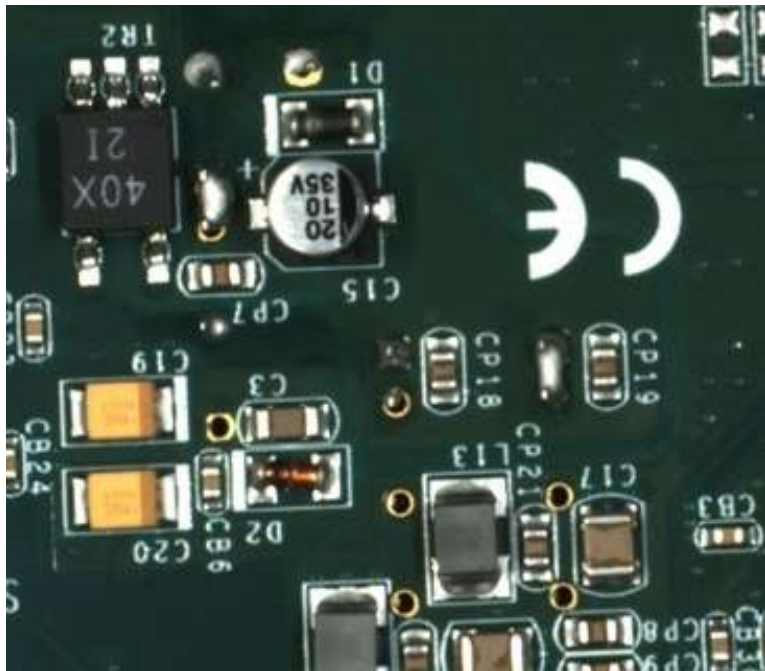
Data Preprocessing - Crop Image

- Individual solder joints remain relatively small compared to the entire board. As a result, defect information occupies only a small portion of the image and may be easily overlooked or weakened during training.
- To improve the visibility and recognizability of defects in the image, each background-removed PCB image is cropped from the top-left to the bottom-right into multiple patches of size 256×256 pixels.
- To prevent defect regions from being partially cut off at the edges of cropped patches, we introduce an overlapping cropping strategy. During the cropping process, each patch is shifted by a certain distance toward the top-left direction in both horizontal and vertical axes.

Data Preprocessing



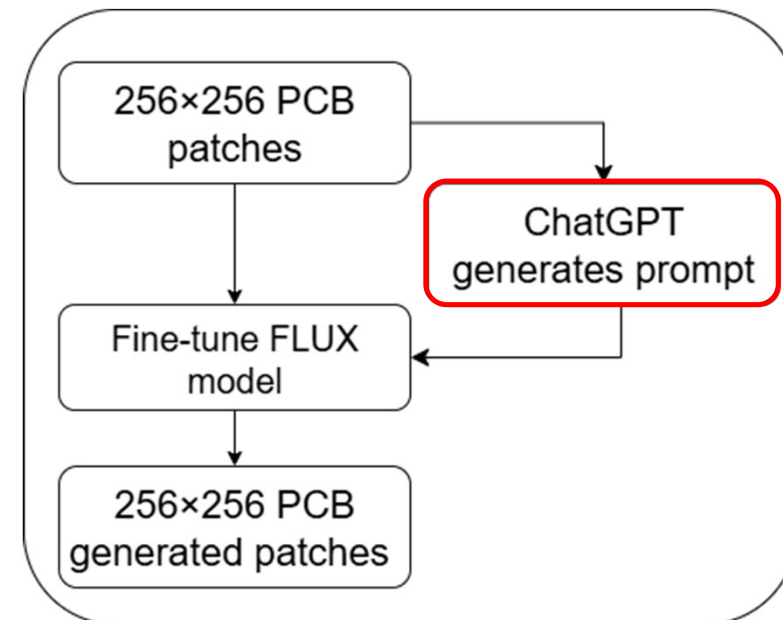
Data Preprocessing - Crop Image



Data Augmentation - ChatGPT generates prompt

- To obtain prompts that accurately reflect the characteristics of each defect type, the defect images were input into ChatGPT along with descriptions of the defect category and its visual features. Special emphasis was placed on highlighting the characteristics of the defect region and its contrast with surrounding normal solder joints, such as differences in reflectivity or shape.
- A dedicated prompt was generated for each defect type.

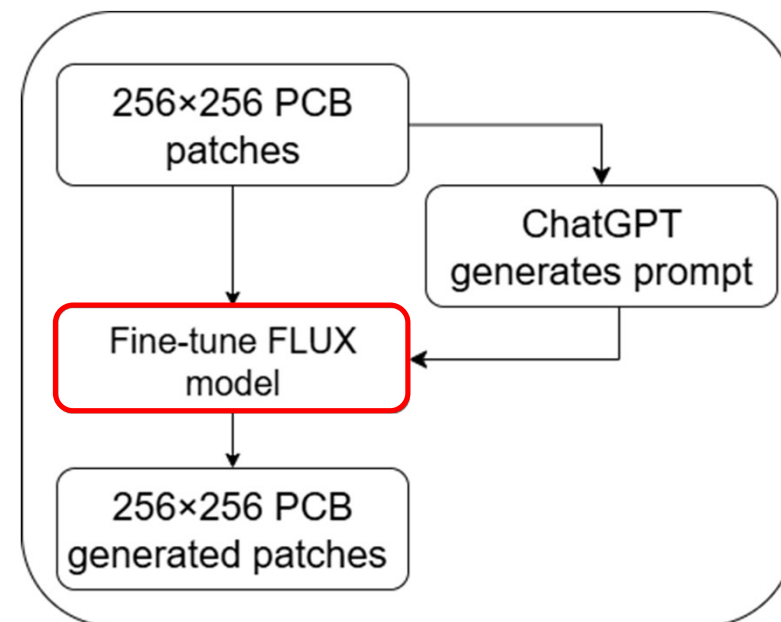
Data Augmentation



Data Augmentation - Fine-tune FLUX model

- To prevent **interference between different defect types** during model fine-tuning, which could degrade generation quality, our method fine-tunes the model separately for each defect category.
- The generative model used in this study is FLUX^[1], specifically the FLUX.1 [dev] version. FLUX is a type of Latent Diffusion Model.

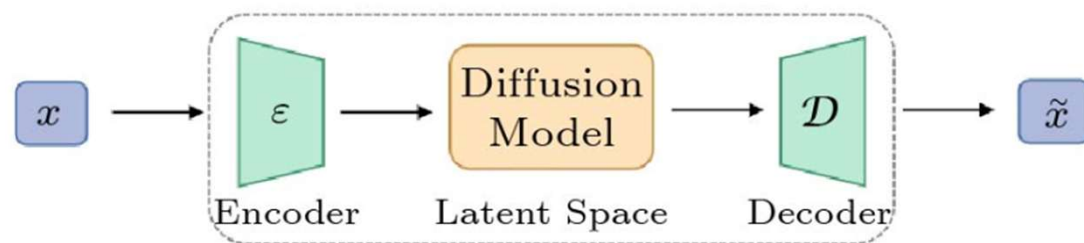
Data Augmentation



[1] B. F. Labs, "FLUX.1 [dev]." <https://huggingface.co/black-forest-labs/FLUX.1-dev>, 2024. Huggingface repository, accessed: 2025-06-17.

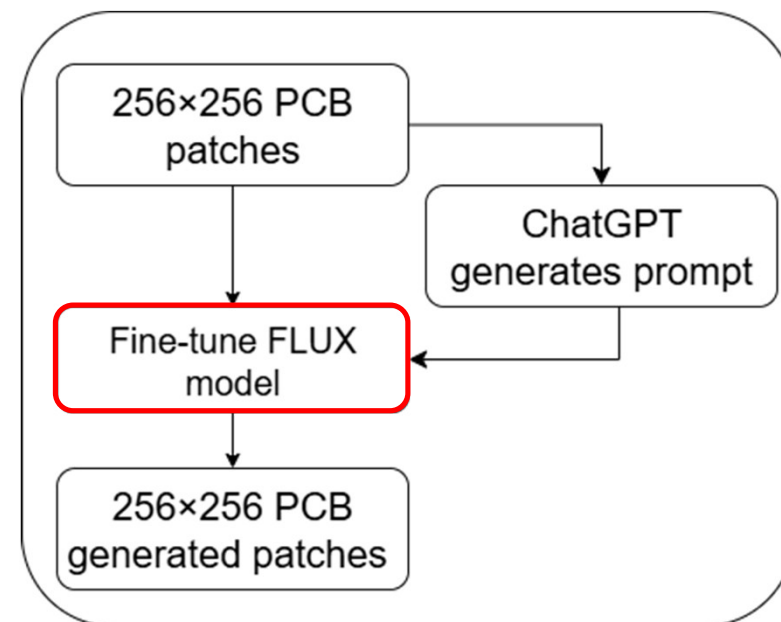
Data Augmentation - Fine-tune FLUX model

- Developed and released by Black Forest Labs in 2024.
- It is a latent diffusion model.
- Cutting-edge output quality.
- Highly competitive prompt adherence.



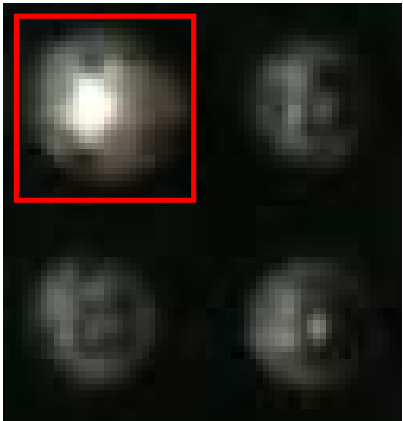
<https://jcst.ict.ac.cn/supplement/1bab19cc-8f20-413f-86ec-84d7acdeb207>

Data Augmentation



Data Augmentation - Fine-tune FLUX model

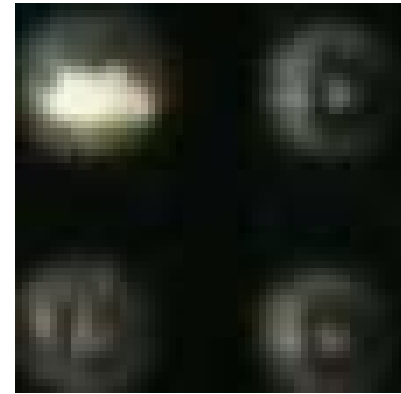
solder_more



original
image



generated
image 1



generated
image 2

Stable Diffusion 3

FLUX

Defect Detection - Fine-tune YOLOv11s model

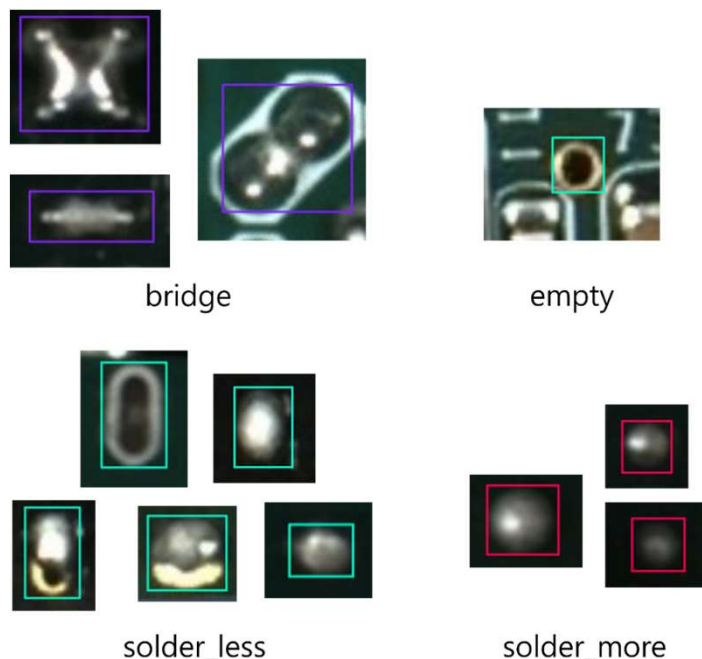
- To **balance both efficiency and accuracy** in automated defect inspection, this study adopts YOLOv11s, which is a version of YOLOv11[1], and YOLOv11 is a model introduced in 2024.
- It replaces the traditional C2f (Cross Stage Partial Bottleneck with two full convolutions) blocks with C3k2 (Cross Stage Partial with kernel size 2) blocks, which reduces the number of parameters and enhances both feature extraction capability and processing speed.
- It also incorporates the C2PSA (Cross Stage Partial with Spatial Attention) module to further improve feature representation and generalization performance.
- The model is fine-tuned using pre-trained weights, enabling effective transfer learning even on limited datasets by leveraging the rich features learned from large-scale datasets.

[1] R. Khanam and M. Hussain, "Yolov11: An overview of the key architectural enhancements," arXiv preprint arXiv:2410.17725, 2024.

Experiments

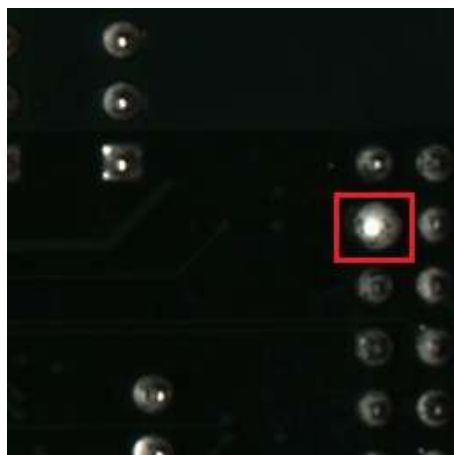
Dataset

- The dataset used in this study consists of non-public PCB images collected from a previous industry-academic collaboration project. The original images have a resolution of 5472×3648 pixels and are full-board PCB photographs that include background information.
- The dataset contains four types of PCB solder joint defects: bridge, empty, solder_less, and solder_more. Each defect type includes a variety of visual appearances and forms.



Dataset

- After applying the preprocessing steps described in the proposed method, the final dataset consists of 256×256 pixel image patches.
- As shown in the table, the solder_less defect is relatively underrepresented in the training set. This **class imbalance** represents one of the key challenges addressed in this study.



defect type	bridge	empty	solder_less	solder_more
training labels	33	22	25	15
test labels	31	11	27	17

Evaluation Metrics - FID

- FID (Fréchet Inception Distance) is a widely used metric in image generation tasks. It is based on the feature space of the Inception network and is designed to quantify the distributional difference between generated and real images.
- By computing the Fréchet distance between the feature vectors of generated and real images, FID **reflects their similarity at a high-level semantic representation**.
- A lower FID score indicates that the generated images are more similar to real ones in terms of feature distribution, thus representing higher quality. Conversely, a higher score implies greater divergence from real data.
- In this study, due to the limited amount of data, FID 64 and FID 192 are used to evaluate the quality of generated images.

$$\text{FID} = \|\mu_r - \mu_g\|_2^2 + \text{Tr} \left(\Sigma_r + \Sigma_g - 2 (\Sigma_r \Sigma_g)^{\frac{1}{2}} \right)$$

Evaluation Metrics - Precision, Recall, and mAP@50

- Precision = $\frac{TP}{TP + FP}$

- Recall = $\frac{TP}{TP + FN}$

- mean Precision = $\frac{1}{N} \sum_{i=1}^N \text{Precision}_i$

- mAP@50 = $\frac{1}{N} \sum_{i=1}^N \text{AP}_i^{\text{IoU}=0.50}$

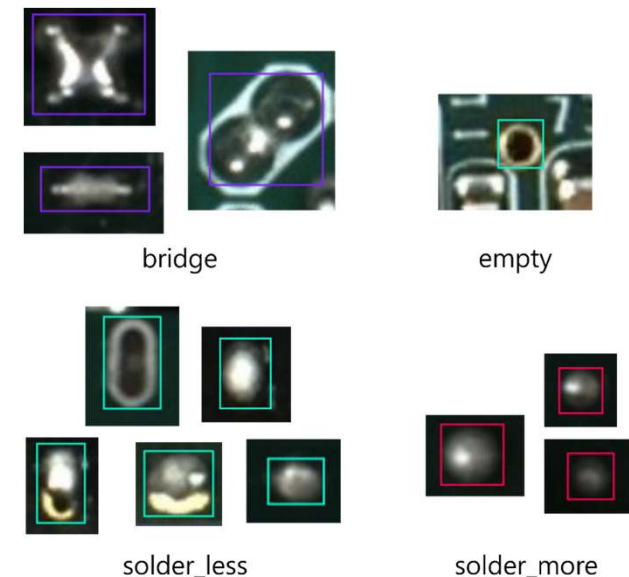
- mean Recall = $\frac{1}{N} \sum_{i=1}^N \text{Recall}_i$

Experiment 1

Objective	To verify whether the proposed data generation method can bring substantial benefits in enhancing defect detection model performance.
Description	Compare the performance of YOLOv11s models fine-tuned with different data sources: Baseline - Fine-tuned using only original data Proposed - Fine-tuned using a combination of original and generated data
Metrics	Precision, Recall, mean Precision, mean Recall, mAP@50

Experiment 1 - ChatGPT generates prompt

defect types	prompt
bridge	A high-resolution close-up of a green printed circuit board (PCB) showing detailed solder joints and components. The focus is on two adjacent solder pads connected by an unintended solder bridge, causing a short circuit . The bridge defect is visually distinct from other solder joints, where solder forms a small, uneven link between two neighboring pads , unlike the smooth, isolated connections of other nearby solder points.
empty	A close-up view of printed circuit boards (PCBs) showcasing soldering defects in high detail. The defects are areas where solder is missing, revealing bare metal pads or through-hole annular rings . These uncovered regions exhibit a distinct lack of the usual smooth, shiny solder coating . The defect areas stand out against the surrounding components and soldered joints, emphasizing the incomplete soldering process. The texture of the PCB surface and the contrast between soldered and unsoldered sections are clearly visible.
solder_less	A high-resolution close-up image of a green printed circuit board (PCB) with multiple soldering pads. Several soldering pads have missing solder defects , appearing as empty or dark holes where solder should be. The defects vary in shape, including oval, circular, and irregular forms , often surrounded by clean metallic pads.
solder_more	A close-up view of a printed circuit board (PCB) with multiple solder joints, showing a defect where one solder joint has excess solder, appearing larger, shinier, and more bulbous compared to the uniformly rounded and smooth surrounding joints.



Experiment 1 - Fine-tune FLUX model

parameter	value
batch_size	2
network_dim	16
network_alpha	16
learning_rate	5×10^{-4}
max_train_steps	1000

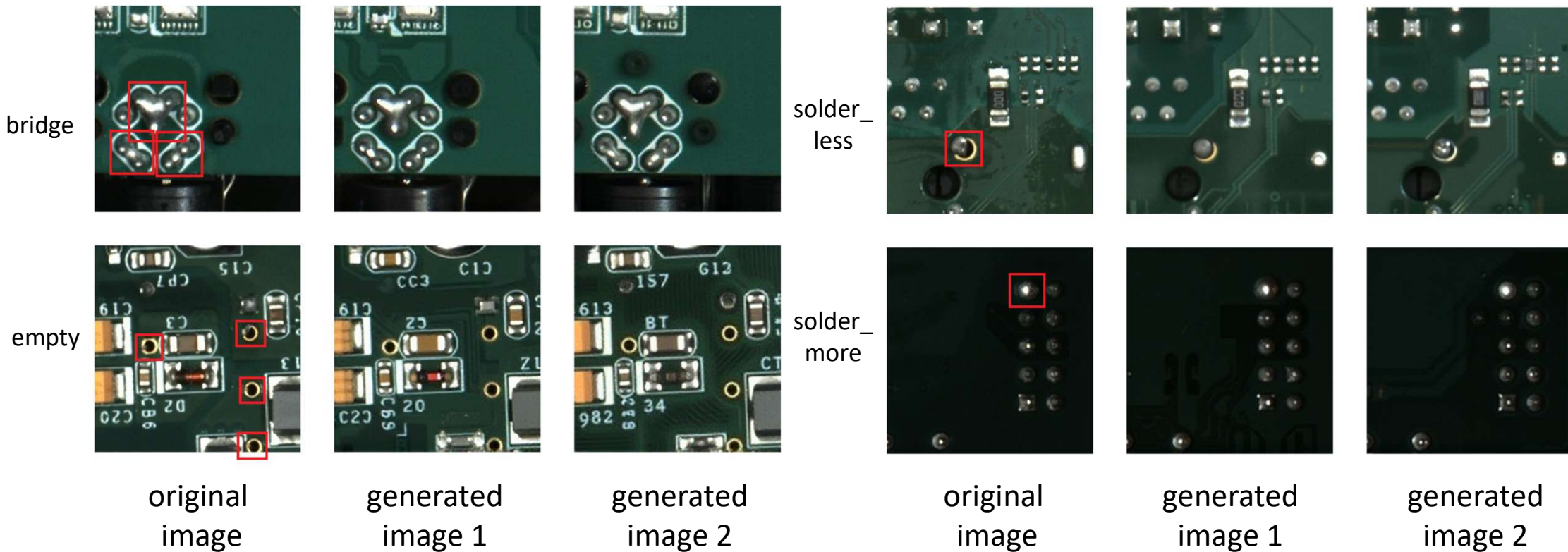
FLUX fine-tuning
parameters

parameter	value
strength_model	0.40
strength_clip	1.00
strength	0.20

FLUX generate
parameters

Experiment 1 - Fine-tune FLUX model

significant variations
in the background



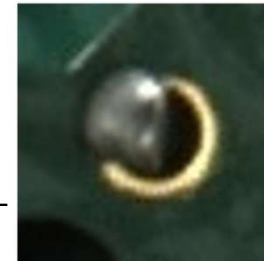
Experiment 1 - Fine-tune FLUX model

noticeable changes in
the shape and details
of the solder joints

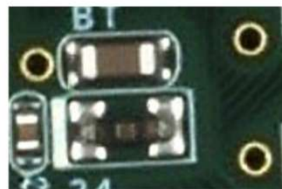
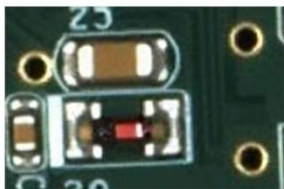
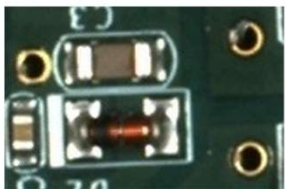
bridge



solder_
less



empty



solder_
more



original
image

generated
image 1

generated
image 2

original
image

generated
image 1

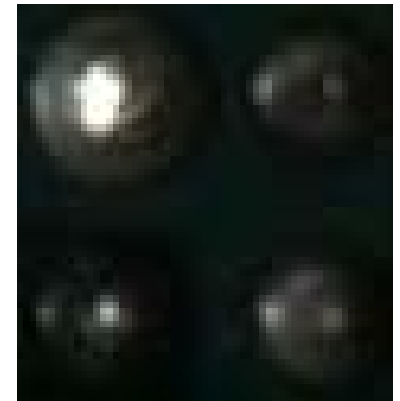
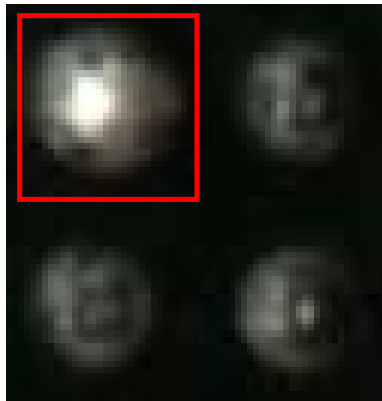
generated
image 2

Experiment 1 - Fine-tune FLUX model

solder_
less



solder_
more



original
image

generated
image 1

generated
image 2

Experiment 1 - Augmented Dataset

	bridge	empty	solder_less	solder_more
original data	32	22	25	15
generated data	51	57	58	67
total	83	79	83	82

Experiment 1 - Result

mAP@50	
w/o generated data	0.936
w/ generated data	0.959
Δ	0.023

Precision	bridge	empty	solder_less	solder_more	mean
w/o generated data	0.879	0.922	0.882	0.932	0.904
w/ generated data	0.903	1	0.912	0.944	0.94
Δ	0.024	0.078	0.03	0.012	0.036

Recall	bridge	empty	solder_less	solder_more	mean
w/o generated data	0.871	0.909	0.828	0.812	0.855
w/ generated data	0.903	0.885	0.815	0.995	0.9
Δ	0.032	-0.024	-0.013	0.183	0.045

Experiment 2

Objective	To compare the proposed method with other data augmentation approaches based on diffusion models, and evaluate whether it outperforms existing SOTA methods in terms of effectiveness.
Description	Compare the effectiveness of different data augmentation methods using both FID scores and the performance of YOLOv11s fine-tuned on mixed data (original + generated). Baseline - YOLOv11s fine-tuned with mixed data generated by the proposed method Proposed - YOLOv11s fine-tuned with mixed data generated by DIAG and CSDG
Metrics	FID, Precision, Recall, mean Precision, mean Recall, mAP@50

Experiment 2 - DIAG

1. Select a defect-free surface image as the base image.
2. Randomly select a defect mask from the defect dataset to **simulate the location and shape** of a defect in a real sample.
3. Input the base image, defect mask, along with the corresponding positive and negative prompts into a Latent Diffusion Model (Stable Diffusion XL) to synthesize defect images.

1. Select a **defect** image as the base image.
2. Select the **corresponding** defect annotation and convert the bounding box into a binary mask.
3. Input the base image, defect mask, the **positive prompt generated by our proposed method**, and **commonly used PCB-specific negative prompts** into a Latent Diffusion Model (Stable Diffusion XL) to synthesize defect images.

positive prompt : “white marks on the wall” and “copper metal scratches”

negative prompt: “smooth, plain, black, dark, shadow”

Experiment 2 - CSDG

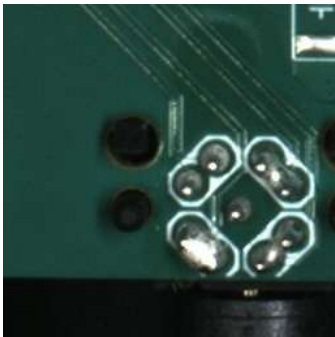
1	loosely cropped	28
2	round-cropped	26
3	dark image	19
4	connections	14
5	left align	12
6	circuit board artifacts	9
7	very grainy image	7
8	interconnections	7
9	close - up of the faces	7
10	metallic reflective	6
11	damaged webcam image	6
12	full photo	6
13	aliasing visible	5
14	little detail	5
15	connector	5
16	simple detail. greenish lighting	5
17	mid closeup	5
18	closeup view	4
19	fig.1	4
20	discord profile picture	4
21	floating symbols	4
22	detailed zoom photo	4
23	cupertino	4

1	a close up of a small electronic device with a disc	3
2	a close up of a green electronic board with a small button	3
3	a close up of a circuit board with a small electronic device	3
4	a close up of a green electronic device with buttons and buttons	2
5	a close up of a circuit board with a few small electronic components	2
6	a close up of a circuit board with many electronic components	2
7	a close up of a computer chip on a motherboard	2
8	a close up of a green electronic board with a yellow button	1
9	a close up of a circuit board with a bunch of buttons	1
10	a close up of a necklace with a chain of beads	1
11	a close up of a green electronic device with a small button	1
12	a close up of a green circuit board with a lot of screws	1
13	a close up of a necklace with a lot of beads on it	1
14	a close up of a computer board with a small white star on it	1
15	a close up of a circuit board with a few small electronics	1
16	a close up of a green electronic device with buttons and a button	1
17	a close up of a black object with a hole in it	1
18	a close up of a circuit board with multiple electronic components	1

prompt: a close up of a green electronic board, loosely cropped, round-cropped, dark image, connections, left align, circuit board artifacts, very grainy image, interconnections, close-up of the faces, metallic reflective, damaged webcam image, full photo, aliasing visible, little detail, connector, simple detail. greenish lighting, mid closeup

Experiment 2

bridge



original
image



generated
image 1



generated
image 2



generated
image 3

Proposed

DIAG

CSDG

Experiment 2

bridge



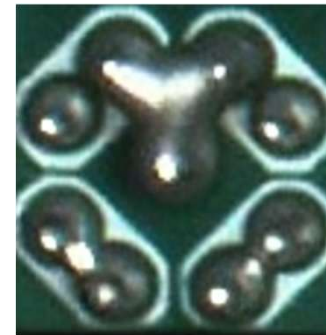
original
image



generated
image 1

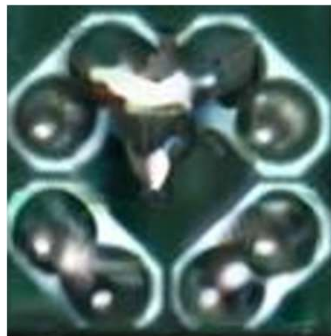


generated
image 2

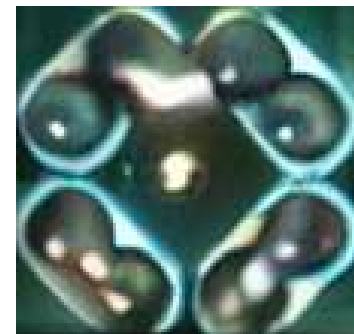
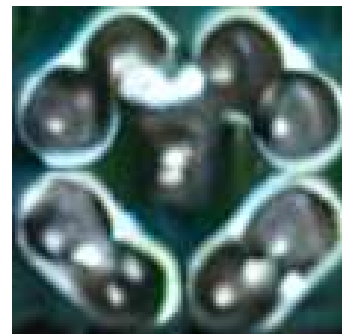


generated
image 3

Proposed



DIAG



CSDG

Experiment 2

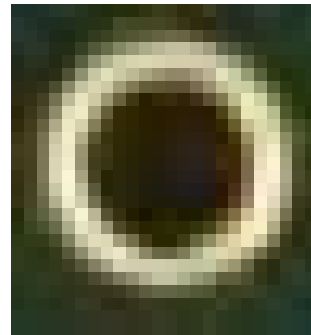
empty



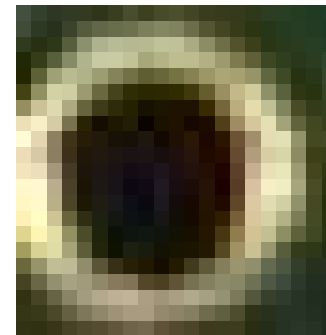
original
image



generated
image 1



generated
image 2



generated
image 3

Proposed

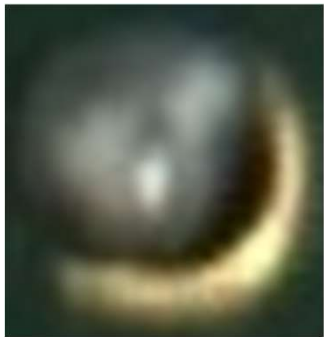
copper rings
deformed

DIAG

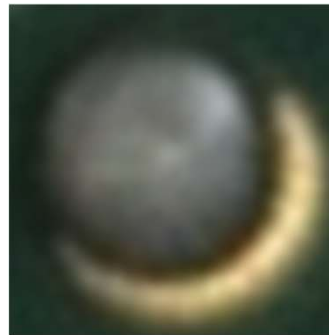
CSDG

Experiment 2

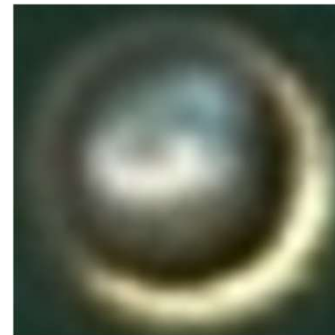
solder_less



original
image



generated
image 1



generated
image 2



generated
image 3

Proposed

copper rings
deformed

DIAG

solder joints
appears blurred
and warped

CSDG

Experiment 2

solder_more

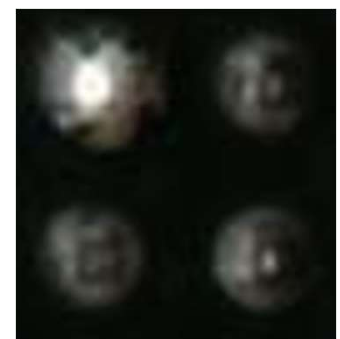
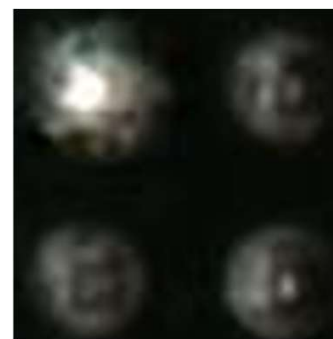
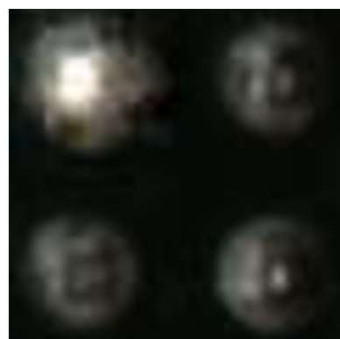


original
image

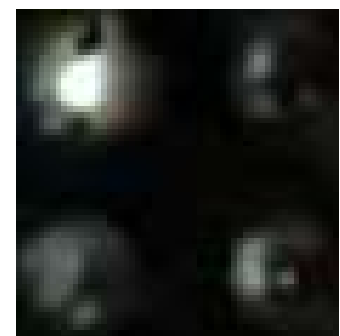
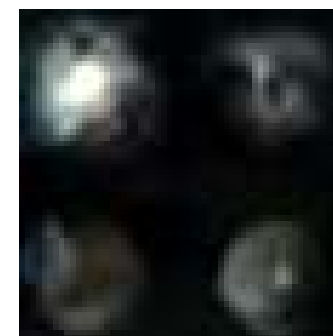
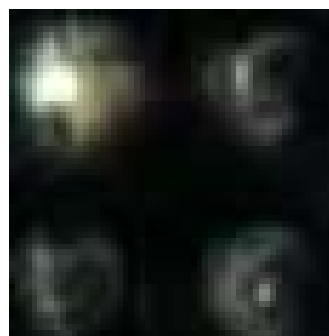


full and solid

Proposed



DIAG



CSDG

generated
image 1

generated
image 2

generated
image 3

Experiment 2 - Result

	FID 64	FID 192
proposed	0.023	0.342
DIAG	0.167	1.408
CSDG	1.058	6.279

Experiment 2 - Result

Precision	bridge	empty	solder_less	solder_more	mean
w/o generated data	0.879	0.922	0.882	0.932	0.904
proposed	0.903	1	0.912	0.944	0.94
DIAG	0.965	0.963	0.8	0.859	0.897
CSDG	0.871	0.938	0.897	0.866	0.893

Recall	bridge	empty	solder_less	solder_more	mean
w/o generated data	0.871	0.909	0.828	0.812	0.855
proposed	0.903	0.885	0.815	0.995	0.9
DIAG	0.888	0.909	0.815	0.765	0.844
CSDG	0.871	0.909	0.704	0.824	0.827

Conclusion

Conclusion

- This study presents a comprehensive solution to the problem of insufficient PCB defect data, covering the entire pipeline from data preprocessing and augmentation to final defect detection.
- In the data generation phase, the proposed method significantly improves detection performance and outperforms two state-of-the-art methods, DIAG and CSDG.
 - The mean Precision increased by 0.036, while DIAG and CSDG saw decreases of 0.007 and 0.011, respectively.
 - The mean Recall improved by 0.045, compared to decreases of 0.011 for DIAG and 0.028 for CSDG.
 - The mAP@50 increased by 0.023, while DIAG and CSDG decreased by 0.013 and 0.027, respectively.
 - In terms of image quality, the proposed method achieved FID 64 = 0.023 and FID 192 = 0.342, significantly better than DIAG (0.167 / 1.408) and CSDG (1.058 / 6.279).
- These results demonstrate that the proposed method not only enhances defect detection accuracy but also generates synthetic data of higher fidelity and distributional similarity to real samples.

Future works

- Although the proposed method is capable of generating data with both diversity and realism, the spatial layout of solder joints and components on the generated PCB images remains largely similar to the reference images in order to preserve authenticity.
- Currently, to ensure that the generated data matches the characteristics of each defect type, it is necessary to manually edit prompts and fine-tune the model separately for each defect category. This process is relatively labor-intensive and complex.
- In the future, developing a unified multi-defect generation framework would significantly streamline the data augmentation process, reduce manual configuration efforts, and enhance practical applicability.